Harun Slahaldin Omar

Software Engineering Student

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Education

- **Master of Engineering, Software Engineering — Concordia University** Jan. 2027 – May 2028
  Co-op Option Expected Graduation: May 2028
- **Bachelor of Engineering, Software Engineering — Concordia University** Sept. 2022 – Present
  Expected Graduation: Fall 2026

Projects

- **Sillage — Event-Driven Multi-Asset Portfolio Engine** *github.com/omar-haha/sillage*
  - Built a portfolio platform in **Python/FastAPI** with a **React/TypeScript** dashboard, using the same decision engine for backtests and live trading across ETFs and crypto.
  - Prevented look-ahead bias by limiting data reads to a known point in time; a blended strategy reached a **0.91 Sharpe** and **19.9% max drawdown** from 2005–2026 after modelled costs.
  - Verified a nine-order IBKR paper lifecycle and built a read-only OpenHands-assisted incident workflow that produces validated diagnoses while keeping recovery owner-approved.
- **MeMantra — Mental Wellness Mobile Application (Capstone)** *github.com/MeMantraa/MeMantra*
  - Co-developed a **React Native/Node.js** mental wellness app with an **11-person Agile team**; the app shipped on the Apple App Store with its API deployed on Render.
  - Built the app's main tab navigation, connected mobile screens to backend APIs, and wrote automated app and API tests with **Jest** and **Supertest**.
  - Led the database migration from Oracle to **PostgreSQL** and contributed to the automated test, code-quality, and security checks used before deployment.
- **Pizza Roma — Restaurant Ordering Platform** *pizzaromamtl.ca*
  - Built and deployed a bilingual **Next.js/TypeScript** ordering platform for an independent Montréal restaurant, covering guided menu configuration, cart, address checks, and staff workflows.
  - Designed server-authoritative pricing and durable **PostgreSQL** orders with immutable price and tax snapshots, retry-safe checkout, and guarded status transitions.
  - Integrated **Helcim** preauthorization with signed-response and amount validation, capturing payment only after staff acceptance; deployed on **OVHcloud** with Docker and Caddy.

Experience

- **Purolator Inc.**, *Delivery Driver* *April 2025 – Present* Laval, QC
  - Handle **200+ daily packages**, including high-value shipments, while maintaining a **99%+ on-time rate**.
  - Resolve routing, access, and labelling issues independently while providing professional customer service.
- **Vanier College**, *Tutor — Calculus I, Calculus II & Web Development* *Aug. 2021 – May 2022* Montréal, QC
  - Tutored **25+ students** through weekly one-on-one and group sessions, helping raise average exam scores by approximately **13%** via personalised learning plans.
  - Developed structured review materials and targeted practice problems for Calculus and intro programming courses.

Programming Skills

Languages: Python, Java, JavaScript / TypeScript, C++, C, C#, SQL
Frameworks & Libraries: React, React Native, Next.js, Node.js, Express, FastAPI, Pandas
Databases & Cloud: PostgreSQL, SQLite, DuckDB, Supabase, AWS (S3, IAM)
Tools & Testing: Git, Docker, Linux, GitHub Actions, Expo EAS, Unity, Jest, pytest
Spoken Languages: English, French

Additional

Interests: Piano, weightlifting, motorcycles, travel (10+ countries)